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What percent of the students at University X are enrolled in a science course but are not enrolled in a biology course?

1. 28 percent of the students at University X are enrolled in a biology course.
2. 70 percent of the students at University X who are enrolled in a science course are enrolled in a biology course.

•

Each Type A machine fills 400 cans per minute, each Type B machine fills 600 cans per minute, and each Type C machine installs 2,400 lids per minute. A lid is installed on each can that is filled and on no can that is not filled. For a particular minute, what is the total number of machines working?

1. A total of 4,800 cans are filled that minute.
2. For that minute, there are 2 Type B machines working for every Type C machine working.

•

In a two-month survey of shoppers, each shopper bought one of two brands of detergent, X or Y, in the first month and again bought one of these brands in the second month. In the survey, 90 percent of the shoppers who bought Brand X in the first month bought Brand X again in the second month, while 60 percent of the shoppers who bought Brand Y in the first month bought Brand Y again in the second month. What percent of the shoppers bought Brand Y in the second month?

1. In the first month, 50 percent of the shoppers bought Brand X.
2. The total number of shoppers surveyed was 5,000.

•

If the total price of n equally priced shares of a certain stock was $12,000, what was the price per share of the stock?

1. If the price per share of the stock had been $1 more, the total price of the n shares would have been $300 more.
2. If the price per share of the stock had been $2 less, the total price of the n shares would have been 5 percent less.

•

In Year X, 8.7 percent of the men in the labor force were unemployed in June compared with 8.4 percent in May. If the number of men in the labor force was the same for both months, how many men were unemployed in June of that year?

1. In May of Year X, the number of unemployed men in the labor force was 3.36 million.
2. In Year X, 120,000 more men in the labor force were unemployed in June than in May.

•

On Monday morning a certain machine ran continuously at a uniform rate to fill a production order. At what time did it completely fill the order that morning?

1. The machine began filling the order at 9:30 a.m.
2. The machine had filled $\frac{1}{2}$ of the order by 10:30 a.m. and $\frac{5}{6}$ of the order by 11:10 a.m.

•

After winning 50 percent of the first 20 games it played, Team A won all of the remaining games it played. What was the total number of games that Team A won?

1. Team A played 25 games altogether.
2. Team A won 60 percent of all the games it played.

•

Michael arranged all his books in a bookcase with 10 books on each shelf and no books left over. After Michael acquired 10 additional books, he arranged all his books in a new bookcase with 12 books on each shelf and no books left over. How many books did Michael have before he acquired the 10 additional books?

1. Before Michael acquired the 10 additional books, he had fewer than 96 books.
2. Before Michael acquired the 10 additional books, he had more than 24 books.

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2. Before Michael acquired the 10 additional books, he had more than 24 books.

•

Last year in a group of 30 businesses, 21 reported a net profit and 15 had investments in foreign markets. How many of the businesses did not report a net profit nor invest in foreign markets last year?

1. Last year 12 of the 30 businesses reported a net profit and had investments in foreign markets.
2. Last year 24 of the 30 businesses reported a net profit or invested in foreign markets, or both.

•

For each landscaping job that takes more than 4 hours, a certain contractor charges a total of r dollars for the first 4 hours plus $0.2r$ dollars for each additional hour or fraction of an hour, where $r > 100$. Did a particular landscaping job take more than 10 hours?

1. The contractor charged a total of $288 for the job.
2. The contractor charged a total of $2.4r$ dollars for the job.

•

If 75 percent of the guests at a certain banquet ordered dessert, what percent of the guests ordered coffee?

1. 60 percent of the guests who ordered dessert also ordered coffee.
2. 90 percent of the guests who ordered coffee also ordered dessert.

•

A tank containing water started to leak. Did the tank contain more than 30 gallons of water when it started to leak? (Note: 1 gallon = 128 ounces)

1. The water leaked from the tank at a constant rate of 6.4 ounces per minute.
2. The tank became empty less than 12 hours after it started to leak.

•

Each of the 45 books on a shelf is written either in English or in Spanish, and each of the books is either a hardcover book or a paperback. If a book is to be selected at random from the books on the shelf, is the probability less than $\frac{1}{2}$ that the book selected will be a paperback written in Spanish?

1. Of the books on the shelf, 30 are paperbacks.
2. Of the books on the shelf, 15 are written in Spanish.

•

A small school has three foreign language classes, one in French, one in Spanish, and one in German. How many of the 34 students enrolled in the Spanish class are also enrolled in the French class?

1. There are 27 students enrolled in the French class, and 49 students enrolled in either the French class, the Spanish class, or both of these classes.
2. One-half of the students enrolled in the Spanish class are enrolled in more than one foreign language class.

•

Last year $\frac{3}{5}$ of the members of a certain club were males. This year the members of the club include all the members from last year plus some new members. Is the fraction of the members of the club who are males greater this year than last year?

1. More than half of the new members are male.
2. The number of members of the club this year is $\frac{6}{5}$ the number of members last year.

•

Machines K, M, and N, each working alone at its constant rate, produce 1 widget in x, y, and 2 minutes, respectively. If Machines K, M, and N work simultaneously at their respective constant rates, does it take them less than 1 hour to

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If Machines K, M, and N work simultaneously at their respective constant rates, does it take them less than 1 hour to produce a total of 50 widgets?

1. $x < 1.5$
2. $y < 1.2$

•

Stations X and Y are connected by two separate, straight, parallel rail lines that are 250 miles long. Train P and train Q simultaneously left Station X and Station Y, respectively, and each train traveled to the other's point of departure. The two trains passed each other after traveling for 2 hours. When the two trains passed, which train was nearer to its destination?

1. At the time when the two trains passed, train P had averaged a speed of 70 miles per hour.
2. Train Q averaged a speed of 55 miles per hour for the entire trip.

•

In a two-story apartment complex, each apartment on the upper floor rents for 75 percent as much as each apartment on the lower floor. If the total monthly rent is $15,300 when rent is collected on all of the apartments, what is the monthly rent on each apartment on the lower floor?

1. An apartment on the lower floor rents for $150 more per month than an apartment on the upper floor.
2. There are 6 more apartments on the upper floor than on the lower floor.

•

A motorboat, which is set to travel at k kilometers per hour in still water, travels directly up and down the center of a straight river so that the change in the boat's speed relative to the shore depends only on the speed and direction of the current. What is the value of k?

1. It takes the same amount of time for the boat to travel 4 kilometers directly downstream as it takes for it to travel 3 kilometers directly upstream.
2. The current flows directly downstream at a constant rate of 2.5 kilometers per hour.

•

If the book value of a certain piece of equipment was $5,000 exactly 5 years ago, what is its present book value?

1. From the time the piece of equipment was purchased, its book value decreased by 10 percent of its purchase price each year of its life.
2. The present book value of another piece of equipment is $2,000.

•

The total cost to charter a bus was shared equally by the people who went on a certain trip. If the total cost to charter the bus was $360, how many people went on the trip?

1. Each person who went on the trip paid $9 to charter the bus.
2. If 4 fewer people had gone on the trip, each person's share of the total cost to charter the bus would have increased by $1.

•

If each of the stamps Carla bought cost 20, 25, or 30 cents and she bought at least one of each denomination, what is the number of 25-cent stamps that she bought?

1. She spent a total of $1.45 for stamps.
2. She bought exactly 6 stamps.

•

A car traveled a distance of d miles in t minutes at an average rate of r miles per minute. What is the ratio of d to r?

1. $t = 30$
2. $d = 25$

•

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Pat is reading a book that has a total of 15 chapters. Has Pat read at least $\frac{1}{3}$ of the pages in the book?

1. Pat has just finished reading the first 5 chapters.
2. Each of the first 3 chapters has more pages than each of the other 12 chapters in the book.

•

Should the choices made by 100 participants in a survey about Product X be said to proportionately represent the preferences of the entire population?

1. In the survey of 100 participants, 70% said they preferred Product X.
2. While the survey provided valuable insights into Product X, the small sample may not be representative of the entire population's preferences.

•

What is a limiting factor when it comes to us using the most effective solution?

1. We have identified several potential solutions, but a lack of funding hinders their implementation.
2. As the most effective solution requires extensive resources, we must opt for a more affordable option given our budget.

•

What are all the important considerations for making accurate projections?

1. Based on the available data, we believe that the market demand for our product will increase by 20% in the next quarter.
2. Without comprehensive market research, our projection may not account for external factors that could influence fluctuations in demand.

•

Does Company X strive to maintain a low defect rate in order to ensure product quality?

1. Company X's manufacturing process has a defect rate of 2% according to quality control data.
2. Monitoring the defect rate in a manufacturing process is essential to ensure product quality, and a low percentage indicates a well-maintained production process.

•

How might career advancement be described as potentially linked to both growth and decline?

1. Working long hours and overtime may lead to higher productivity and career advancement but is likely to cause a decline in physical and mental well-being.
2. Maintaining a work-life balance is important to reduce the negative effect that working too much has on a person's quality of life.

•

What is an example of how investment in healthcare resources might sometimes create a challenge?

1. Investment in both advanced medical technology and specialized treatments can improve patient outcomes for certain conditions. Nevertheless, it may put a strain on healthcare budgets.
2. When it comes to allocating resources in the healthcare sector, directing money toward the purchase of advanced medical equipment could potentially limit access to other essential services and treatments.

•

A person bought x number of apples, y number of oranges, and z number of mangoes from a shop. What is the total price of this transaction if each apple costs \$2, each orange costs \$2.50, and each mango costs \$3?

1. The number of apples bought is 12 more than the number of oranges bought. The number of mangoes bought is half the number of apples bought.
2. The average number of fruits purchased from the three varieties is greater than 29 and less than 33.

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•

Adam received a certain monetary amount in bills having only three denominations: $1, $5, and $10. What is the total monetary amount in bills that he received?

1. The numbers of $1, $5, and $10 bills Adam received, respectively, are in the ratio 3:5:8.
2. The ratio of the monetary amount Adam received in $1 bills to the monetary amount he received in $10 bills is 3:80.

•

Exactly 3 investors—A, B, and C—invested in a certain business, each for a certain number of months. For these investments, what is the ratio of the profit earned by A to the profit earned by B to the profit earned by C? (Note: Profit earned is proportional to the product of investment amount and investment duration.)

1. A invested twice as much as B, and C invested half as much as A and B together.
2. All 3 investments were for the same number of months.

•

The ratio of A's current age to B's current age is 5:8. What will be the sum of their ages 10 years from now? (Note: Assume that neither A nor B dies in the next 10 years.)

1. If A had been born 2 years later and B had been born 2 years earlier, then the ratio of A's current age to B's current age would be 14:25.
2. A's age 5 years ago was half the age that B will be 2 years from now.

•

A's father is 6 years older than her mother. The ratio of A's age to her mother's age is 1:4. What is the age of A's father?

1. A's father is 30 years older than A.
2. The ratio of A's age to her father's age is 4:19.

•

The salary of A is what percentage of the combined salaries of B and C?

1. Twice the salary of A is equal to three times the salary of B.
2. 150% of B's salary is $160 more than 120% of C's salary.

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